SHANKHA DEEP SWARNAKAR

Computer Science & Engineering Student | AI & Computer Vision | Full-Stack Development
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Portfolio | GitHub

PROFESSIONAL SUMMARY

Computer Science Engineering student specializing in Artificial Intelligence, Computer Vision, and Full-Stack Web Development. Delivered 3+ end-to-end AI and web projects using Python, React.js, and Firebase, with emphasis on real-time systems and human-computer interaction. Completed 5 virtual internships spanning AI automation, data analytics, and algorithm design — earning Grade A or above in all programs. Seeking a software engineering or research internship to apply machine learning and full-stack development skills in a product-driven environment.

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C, JavaScript, SQL, HTML5, CSS3
- **Frameworks & Libraries:** React.js, OpenCV, MediaPipe, Node.js, Express.js
- **Tools & Databases:** Git, GitHub, Firebase Firestore, MongoDB, PyAutoGUI
- **Core Competencies:** Computer Vision, Machine Learning, Data Analytics, Human-Computer Interaction, REST APIs, Responsive Web Design
- **Design & Multimedia:** Adobe Premiere Pro, ZBrush, Lightroom, Canva

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science and Engineering *Expected: May 2027*
Techno India University, Kolkata, West Bengal

- Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Compiler Design, Computer Networks

INTERNSHIPS & VIRTUAL EXPERIENCE

Data Analytics Job Simulation | Deloitte via Forage *May 2026*

- Completed 2 practical consulting modules covering data analysis and forensic technology within a simulated enterprise environment.
- Cleaned and analyzed datasets to generate business intelligence reports, reducing decision ambiguity across 3 simulated client scenarios.
- Applied data visualization techniques to effectively communicate insights to non-technical stakeholders.

AI Automation, Design and Scaling Software | Techno Billion AI *January 2026*

- Completed virtual internship covering AI-driven automation pipelines, software design principles, and production-level scaling strategies — Grade A+.
- Demonstrated strong command of automation architecture and consistent task delivery across all program modules.

Certified AI Professional (C-CAP) | Techno Billion AI & CoRover.AI *January 2026*

- Earned the C-CAP credential validating applied AI skills in conversational AI, intelligent automation, and production AI system design — Grade A.

Applied AI: Data, Web and Documentation | Techno Billion AI *February 2026*

- Delivered AI-integrated solutions across 3 domains: data processing pipelines, web application integration, and structured technical documentation — Grade A.
- Bridged research-level AI concepts with practical software engineering workflows.

Dynamic Programming: Concepts and Fibonacci Series | Techno Billion AI *January 2026*

- Mastered memoization and tabulation techniques, achieving up to 40% improvement in time complexity over naive approaches — Grade A+.
- Implemented and analyzed Fibonacci-based algorithmic patterns with thorough complexity analysis.

PROJECTS

Intelligent Gesture-Controlled Desktop Assistant *2025*

Technologies: Python, OpenCV, MediaPipe, PyAutoGUI

- Built an AI-powered touchless desktop controller that translates real-time hand gestures into OS-level commands including cursor movement, clicking, dragging, and scrolling.
- Optimized hand landmark tracking algorithms to reduce gesture recognition latency, achieving stable performance across variable lighting conditions.
- Mapped 6+ distinct gestures to system commands using MediaPipe's 21-point hand landmark model integrated with PyAutoGUI.

Personal Portfolio & Creative Web Showcase *2025*

Technologies: JavaScript, HTML5, CSS3, Vercel CI/CD

- Developed a performance-optimized portfolio showcasing 3 major projects, digital artwork, and a resume — with fast load times across all devices.
- Deployed via Vercel CI/CD pipeline at shankha-deep-swarnakar.vercel.app for seamless continuous delivery.

PrepAI Pro – AI-Powered Interview Preparation & Career Readiness Platform (Project link) *2025*

Technologies: Python Flask, MySQL, HTML5, CSS3, Bootstrap, JavaScript, Gemini AI, Chart.js, SpeechRecognition, PyPDF2

- Developed a full-stack web application using Python Flask, MySQL, JavaScript, and Google Gemini AI to support interview preparation and career development.
- Implemented AI-powered mock interviews, ATS resume analysis, voice-based communication assessment, aptitude testing, and personalized study plan generation.
- Built secure authentication, role-based access control, admin dashboard, and candidate management functionalities.
- Integrated speech-to-text processing, AI-driven feedback generation, performance analytics dashboards, and career recommendation systems.

ResumeForge – AI-Assisted ATS Resume Builder (Project link) *2025*

Technologies: React.js, Tailwind CSS, JavaScript, jsPDF, Local Storage, HTML5, CSS3

- Developed a modern resume-building platform that enables students, job seekers, and freelancers to create ATS-optimized resumes with real-time editing and instant PDF generation.
- Implemented an ATS evaluation system that analyzes resume structure, content completeness, skills, projects, and professional experience to generate actionable improvement recommendations.
- Built a live resume rendering engine with dynamic form management, multi-template support, and responsive preview functionality for professional document creation.
- Designed and developed customizable resume templates following ATS-compliant formatting standards to improve compatibility with modern recruitment systems.
- Integrated automated draft saving, profile completion tracking, and user-friendly validation mechanisms to enhance productivity and user engagement.
- Leveraged React Hooks and component-based architecture to create a scalable, maintainable, and high-performance single-page application.

- Deployed a fully functional web application featuring resume creation, ATS scoring, PDF export, responsive design, and persistent local storage support.

WORK EXPERIENCE

Software Development Engineering (Web) Intern | Hunt Digital Media, Virtual *July 2026 – Present*

- Designed and developed a SIM Operations Management System, including SIM Inventory, Slot Management, SIM Allocation, and SIM Rotation modules.
- Conceptualized and documented the complete system architecture, database design, allocation workflow, and technical SRS for scalable operations.
- Built responsive admin dashboards with real-time monitoring, SIM health tracking, slot visualization, and automated allocation workflows.
- Collaborated with the tech team to define backend APIs, database relationships, and business logic for efficient SIM lifecycle management.

Freelance Social Media Manager & Content Creator | Remote *October 2025 – Present*

- Managed digital content strategies for 3+ clients, improving audience engagement through data-driven posting schedules and brand-aligned creative assets.
- Analyzed weekly platform metrics to refine content style, increasing click-through rates and organic reach across Instagram and LinkedIn.
- Produced promotional videos and visual assets using Adobe Premiere Pro and ZBrush, delivering polished drafts within 48-hour turnaround.

CERTIFICATIONS

- **Data Analytics Job Simulation —** Deloitte via Forage (May 2026)
- **AI Automation, Design and Scaling Software (Grade A+) —** Techno Billion AI (January 2026)
- **Certified AI Professional — C-CAP (Grade A) —** Techno Billion AI & CoRover.AI (January 2026)
- **Applied AI: Data, Web and Documentation (Grade A) —** Techno Billion AI (February 2026)
- **Dynamic Programming: Concepts and Fibonacci Series (Grade A+) —** Techno Billion AI (January 2026)
- **Full-Stack Web Development and Responsive Design —** Self-Learning & Implementation (2025)

ACHIEVEMENTS & ACTIVITIES

- Delivered 3 end-to-end technical projects spanning AI, computer vision, and full-stack web development — each deployed or demonstrated in a real-world context.
- Completed 5 virtual internships across AI automation, data analytics, and algorithms in under 3 months — earning Grade A or above in all programs.

Active contributor to campus technical and cultural events, combining software development expertise with creative media production